Set 18: Product Inventory Management

DSA – Data Handling and Visualization | Lab Experiment

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Aim

To analyse product inventory using a bar chart of quantities, a stacked bar chart by product category and a labelled table, and to combine them in an interactive Tableau dashboard.

Software / Tools Used

R 4.3.3 with ggplot2 (charts), gridExtra (tables) and patchwork (dashboard layout); Tableau Desktop / Tableau Public (interactive dashboard).

Dataset

Product ID	Product Name	Quantity Available	Price (in $)
1	Product A	250	20
2	Product B	175	15
3	Product C	300	18

Assumed product categories (not given in the dataset):

Product Name	Category (assumed)
Product A	Category 1
Product B	Category 2
Product C	Category 1

Assumption: The dataset has no product category column, so the categories above were assumed to answer Q2. Replace them with the real categories in the data frame if available.

Program – Dataset Creation (common to all R tasks)

```
library(ggplot2)
library(gridExtra)
library(patchwork)
options(scipen = 999)
# Plain white table theme with black borders (no colours)
tbl_theme <- ttheme_minimal(
  core    = list(fg_params = list(col = "black", fontsize = 11),
                 bg_params = list(fill = "white", col = "black")),
  colhead = list(fg_params = list(col = "black", fontface = "bold", fontsize = 11.5),
                 bg_params = list(fill = "white", col = "black")))
inv <- data.frame(
  Product_ID   = c(1, 2, 3),
  Product_Name = c("Product A", "Product B", "Product C"),
  Quantity     = c(250, 175, 300),
  Price        = c(20, 15, 18),
  # ASSUMED: the dataset has no category column, so categories are assumed
  Category     = c("Category 1", "Category 2", "Category 1")
)
write.csv(inv, "Product_Inventory.csv", row.names = FALSE)  # for Tableau
pal <- c("Product A" = "#0072B2", "Product B" = "#E69F00", "Product C" = "#009E73")
```

Q1. Bar Chart – Quantity of Each Product

> *Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.*

R Program:

```
print(inv[, c("Product_Name", "Quantity")], row.names = FALSE)
cat("Total units in stock:", sum(inv$Quantity), "\n")
p_bar <- ggplot(inv, aes(x = Product_Name, y = Quantity, fill = Product_Name)) +
  geom_col(width = 0.55) + geom_text(aes(label = Quantity), vjust = -0.5, size = 4.5) +
  scale_fill_manual(values = pal) + scale_y_continuous(limits = c(0, 350)) +
  labs(title = "Quantity of Each Product in Inventory", x = "Product", y = "Quantity
Available") +
  theme_minimal(base_size = 12) + theme(plot.title = element_text(face = "bold", hjust = 0.5),
legend.position = "none")
print(p_bar)
ggsave("s18_q1.png", p_bar, width = 7, height = 4.5, dpi = 150, bg = "white")
```

Console Output:

```
 Product_Name Quantity
    Product A      250
    Product B      175
    Product C      300
Total units in stock: 725
```

Output:

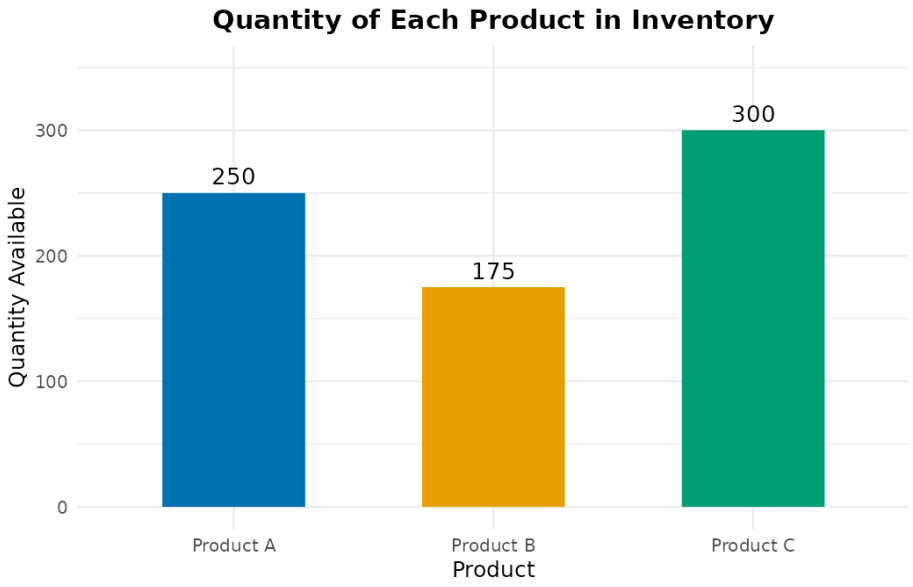


Figure 1: Quantity available per product

Inference:

Product C has the highest stock (300 units), followed by Product A (250) and Product B (175, the lowest). Total inventory is 725 units.

Q2. Stacked Bar Chart – Quantity within Product Categories

> *Generate a stacked bar chart to show the quantity of each product within different product categories.*

R Program:

```
print(xtabs(Quantity ~ Category + Product_Name, data = inv))
p_stk <- ggplot(inv, aes(x = Category, y = Quantity, fill = Product_Name)) +
  geom_col(width = 0.55, colour = "white") +
  geom_text(aes(label = paste0(Product_Name, ": ", Quantity)), position = position_stack(vjust
= 0.5),
            colour = "white", size = 3.8, fontface = "bold") +
  scale_fill_manual(values = pal) +
  labs(title = "Quantity of Each Product within Product Categories", x = "Product Category", y
= "Quantity Available", fill = "Product") +
  theme_minimal(base_size = 12) + theme(plot.title = element_text(face = "bold", hjust = 0.5))
print(p_stk)
ggsave("s18_q2.png", p_stk, width = 7, height = 4.5, dpi = 150, bg = "white")
```

Console Output:

```
            Product_Name
Category     Product A Product B Product C
  Category 1       250         0       300
  Category 2         0       175         0
```

Output:

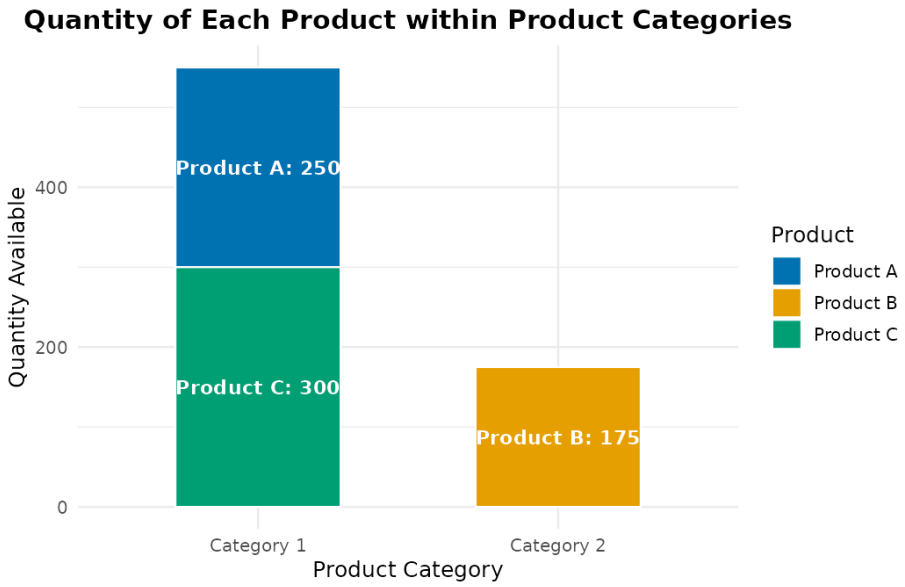


Figure 2: Quantity of each product within assumed categories

Inference:

Category 1 holds 550 units (75.9%), made up of Product C (300) and Product A (250). Category 2 holds only Product B with 175 units (24.1%). Stock is concentrated in Category 1.

Q3. Table – Inventory Data for Each Product

> *Build a table to show the inventory data for each product. Label the table elements.*

R Program:

```
tbl <- inv[, c("Product_ID", "Product_Name", "Quantity", "Price")]
tbl$Inventory_Value <- inv$Quantity * inv$Price
names(tbl) <- c("Product ID", "Product Name", "Quantity Available", "Price (in $)", "Inventory
Value (in $)")
print(tbl, row.names = FALSE)
tg <- tableGrob(tbl, rows = NULL, theme = tbl_theme)
png("s18_q3.png", width = 1200, height = 260, res = 150)
```

```
grid.arrange(grid::textGrob("Inventory Data by Product", gp = grid::gpar(fontsize = 14,
fontface = "bold")),
             tg, ncol = 1, heights = c(0.3, 1))
invisible(dev.off())
```

Console Output:

```
 Product ID Product Name Quantity Available Price (in $) Inventory Value (in $)
          1    Product A                250           20                    5000
          2    Product B                175           15                    2625
          3    Product C                300           18                    5400
```

Table Output:

Product ID	Product Name	Quantity Available	Price (in $)	Inventory Value (in $)
1	Product A	250	20	5,000
2	Product B	175	15	2,625
3	Product C	300	18	5,400
	Total	725		13,025

Inference:

Inventory value (Quantity × Price) is highest for Product C ($5,400) and totals $13,025. Product B holds the least stock value ($2,625).

Q4. Tableau Dashboard – Interactive Exploration

> *Develop a Tableau dashboard combining the bar chart, stacked bar chart, and the table for interactive exploration of inventory data.*

R Program:

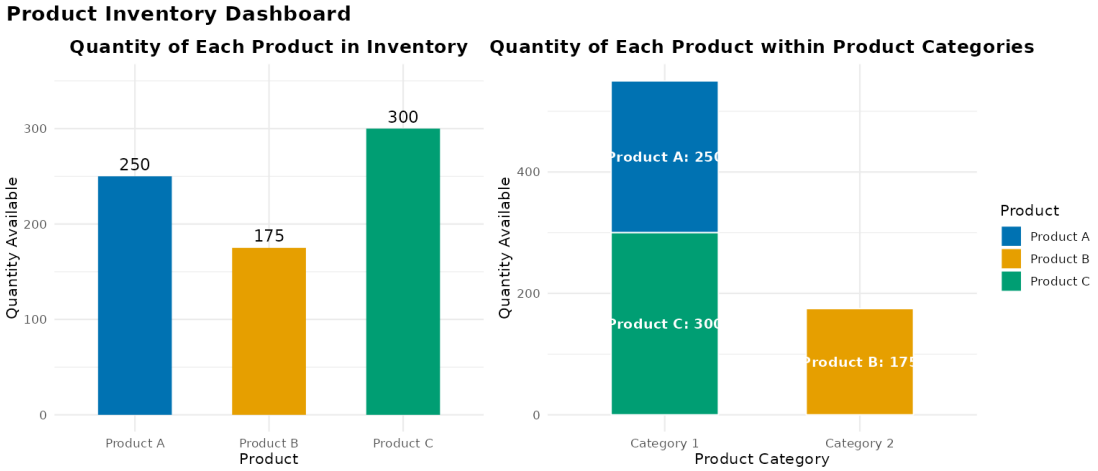
```
dash <- (p_bar | p_stk) / wrap_elements(full = tg) + plot_layout(heights = c(1.4, 0.6)) +
  plot_annotation(title = "Product Inventory Dashboard",
                  theme = theme(plot.title = element_text(face = "bold", size = 16)))
ggsave("s18_dash.png", dash, width = 12, height = 7, dpi = 130, bg = "white")
```

Procedure in Tableau:

1. Open Tableau → **Connect → Text file** → **Product_Inventory.csv**.
2. **Sheet 1 – "Quantity by Product":** drag Product Name to Columns and SUM(Quantity) to Rows. Drag Product Name to Color and show labels.
3. **Sheet 2 – "Stacked by Category":** drag Category to Columns and SUM(Quantity) to Rows, then drag Product Name to **Color** (bars stack automatically) and to Label.
4. **Sheet 3 – "Inventory Table":** drag Product ID, Product Name, Quantity and Price to Rows/Text. Add a calculated field **Inventory Value = [Quantity] * [Price]**. Rename the headers.
5. From the menu choose **Dashboard → New Dashboard** and set **Size → Automatic** (or Fixed 1200 × 800).
6. Drag the sheets onto the canvas: Quantity by Product (top-left), Stacked by Category (top-right), Inventory Table (bottom). Add a **Text** object at the top as the dashboard title.
7. On each chart, open the sheet's drop-down and select **Use as Filter**, so clicking a bar, point or slice filters the other views.
8. Go to **Dashboard → Actions → Add Action → Highlight** (Run on: Hover) so related marks highlight across all views.

9. Right-click each quick filter → **Apply to Worksheets → All Using This Data Source** (Product_Inventory) so one filter controls every view.
10. Save as a packaged workbook (**File → Save As → .twbx**) or publish to Tableau Public.

Dashboard Layout:



Product ID	Product Name	Quantity Available	Price (in $)	Inventory Value (in $)
1	Product A	250	20	5000
2	Product B	175	15	2625
3	Product C	300	18	5400

Reference layout of the dashboard (drawn in R). The interactive version is built in Tableau using the procedure above.

Inference:

Selecting a category in the stacked bar filters the bar chart and table to the products in that category, so stock levels can be compared group by group.

Result

The product inventory was visualised with a bar chart, a stacked bar chart and a labelled table in R, and combined into an interactive Tableau dashboard. Product C has the most stock (300 units), and Category 1 (assumed) holds 75.9% of the 725 units.